

question paper contains 8 printed pages]

Roll No. :

. of Q. Paper : 611 I

e Paper Code : 32357505

of the Course : B.Sc.(Hons.)
Mathematics : DSE-I

of the Paper : Discrete Mathematics

ster : V

: 3 Hours Maximum Marks : 75

ctions for Candidates :

-) Write your Roll No. on the top immediately on receipt of this question paper.
-) Do any **two** parts from each question.

Section - I

-) Define covering relation in an ordered set. Prove that if X is any set, then in the ordered set $\wp(X)$ equipped with the set inclusion relation given by $A \leq B$ if and only if $A \subseteq B$ for all $A, B \in \wp(X)$, a subset B of X covers a subset A of X if and only if $B = A \cup \{b\}$, for some $b \in X \sim A$.

6

P.T.O.

(b) Let $\mathbb{N}_0$ be the set of whole numbers equipped with the partial order $\leq$ defined by $m \leq n$ if and only if m divides n and let $\wp(\mathbb{N})$ be the power set of $\mathbb{N}$ equipped with the partial order given by $A \leq B$ if and only if $A \subseteq B$ for all $A, B \in \wp(\mathbb{N})$. In which of the following cases is the map $\varphi : P \rightarrow Q$ order-preserving?

(i) $P = Q = \mathbb{N}_0$ and $\varphi(x) = nx \forall x \in P$, where $n \in \mathbb{N}_0$ is fixed.

(ii) $P = Q = \wp(\mathbb{N})$ and φ defined by

$$\varphi(A) = \begin{cases} \{1\} & \text{if } 1 \in A \\ \{2\} & \text{if } 2 \in A \text{ but } 1 \notin A \\ \emptyset & \text{otherwise} \end{cases}$$

(c) Let $P = \{a, b, c, d, e, f, u, v\}$. Draw a diagram of the ordered set $(P, \leq)$ where

$$v < a < c < d < e < u, \quad a < f < u, \\ v < b < c, \quad b < f$$

Also, find out $a \vee b$, $a \wedge b$, $e \vee f$ and $e \wedge f$.

- (a) Let V be a vector space and let $M = \text{Sub } V$, the set of all subspaces of V . Prove that $(M, \subseteq)$ is a lattice as an ordered set but is not a sublattice of the lattice $(L, \subseteq)$, where $L = \wp(V)$, the power set of V . 6.5
- (b) Prove that in a lattice L , the following inequalities are satisfied :
- (i) $a \wedge (b \vee c) \geq (a \wedge b) \vee (a \wedge c) \quad \forall a, b, c \in L$ 3
- (ii) $(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) \leq (a \vee b) \wedge (b \vee c) \wedge (c \vee a) \quad \forall a, b, c \in L$ 3.5
- (c) Let $(L, \leq)$ be a lattice as an ordered set. Define two binary operations $+$ and $\cdot$ on L by $x+y = x \vee y = \sup \{x, y\}$ and $x \cdot y = x \wedge y = \inf \{x, y\}$. Prove that $(L, +, \cdot)$ is an algebraic lattice. 6.5

Section - II

- (a) Define a distributive lattice. Prove that a homomorphic image of a distributive lattice is distributive. 6

- (b) Use the Quine-McCluskey method to find minimal form of :

$$xyz' + xy'z + xy'z' + x'yz + x'y'z$$

- (c) (i) Find the conjunctive normal form of :

$$(x_1 + x_2 + x_3)(x_1x_2 + x_1'x_3)'$$

- (ii) Find the disjunctive normal form of :

$$x_1'x_2 + x_3(x_1' + x_2)$$

4. (a) (i) Prove that $(x \wedge y)' = x' \vee y'$ and

$(x \vee y)' = x' \wedge y'$ for all x, y in a Boolean algebra B. 5. 3.

- (ii) Show that the lattice $(\{1, 2, 4, 5, 10, 20, \text{gcd, lcm}\})$ does not form a Boolean algebra for the set of positive divisor of 20.

- (b) Using the Karnaugh Diagrams, find minimum form for p and q where :

$$p = (x_1 + x_2)(x_1 + x_3) + x_1x_2x_3$$

$$q = x_1x_2x_3 + x_1x_2'x_3' + x_1'x_2x_3 + x_1'x_2'x_3 + x_1'x_2'x_3'$$

- (c) Draw the contact diagram and give the symbolic representation (using seven gates) of the circuit given by

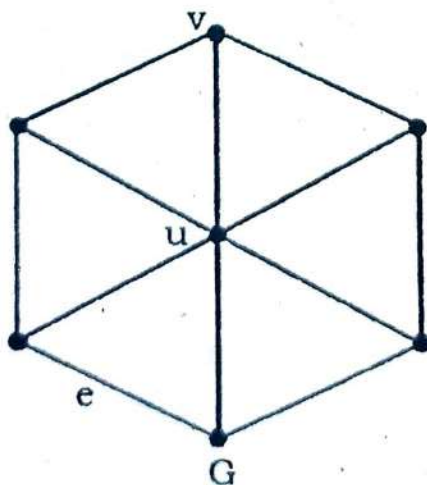
$$p = (x_1 + x_2 + x_3)(x_1' + x_2)(x_1x_3 + x_1'x_2)(x_2' + x_3)$$

6.5

Section - III

5. (a) (i) Draw pictures of the subgraphs $G \setminus \{e\}$, $G \setminus \{v\}$ and $G \setminus \{u\}$ of the following graph G .

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- (ii) Answer the Königsberg bridge problem and explain your answer with graph.

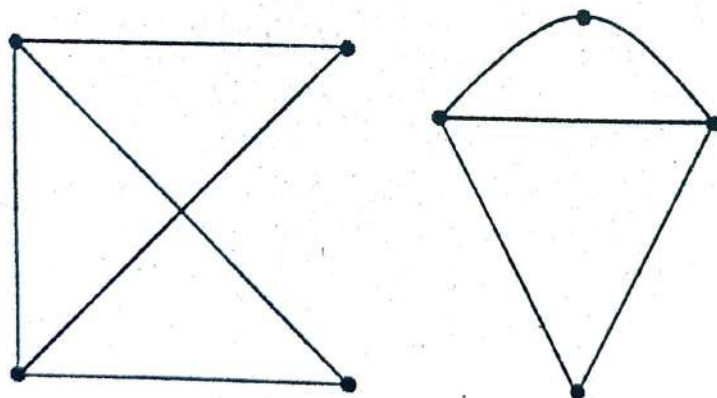
3

- (b) (i) Draw K_4 and $K_{3,4}$.

3

- (ii) For the below pair of graphs, either label the graphs so as to exhibit an isomorphism or explain why graphs are not isomorphic.

3



- (c) (i) Does there exist a graph G with 28 edges and 12 vertices, each of degree 3 or 4. Justify your answer.

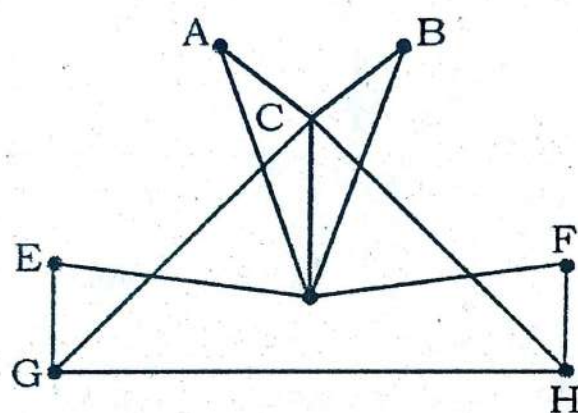
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- (ii) A complete graph with more than two vertices is not bipartite. Justify this statement.

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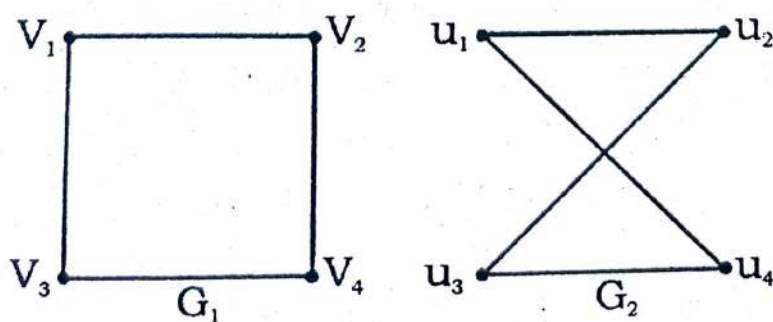
- (iii) Draw a graph whose degree sequence is 1,1,1,1,1,1. 2

6. (a) Consider the Graph G given below. Is it Hamiltonian? Is it Eulerian? Explain your answers. 6.5



- (b) Find the adjacency matrices A_1 and A_2 of the graphs G_1 and G_2 shown below. Find a permutation matrix P such that $A_2 = PA_1P^T$.

6.5



- (c) Apply the improved version of Dijkstra's Algorithm to find a shortest path from A to Z. Write steps. 6.5

